Secure Messaging Web Application with ECDH & ChaCha20

User Guide

1. Running the program

1. Unzip file

2. Open Terminal

3. Install the following:
   pip install cryptography Flask jsonify

4. Navigate to /S1-A-Application

5. Run app.py file (from IDE or terminal)

6. Visit Link in browser
   http://127.0.0.1:8011/

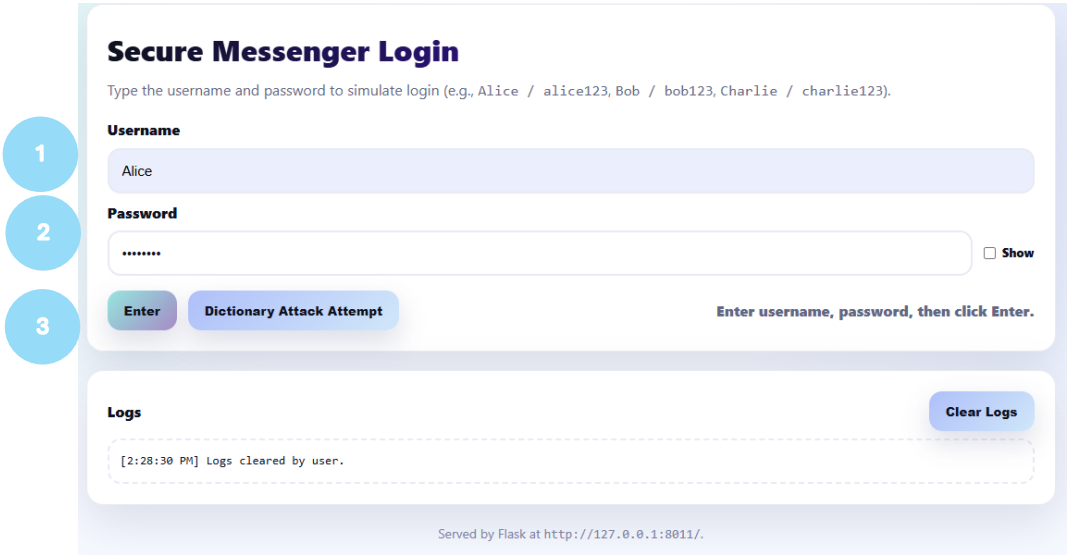
Alternatively: open ICS 344 project in your IDE and run the app.py file.
(WebStorm was used during development)

2. Testing the Program

1. Login

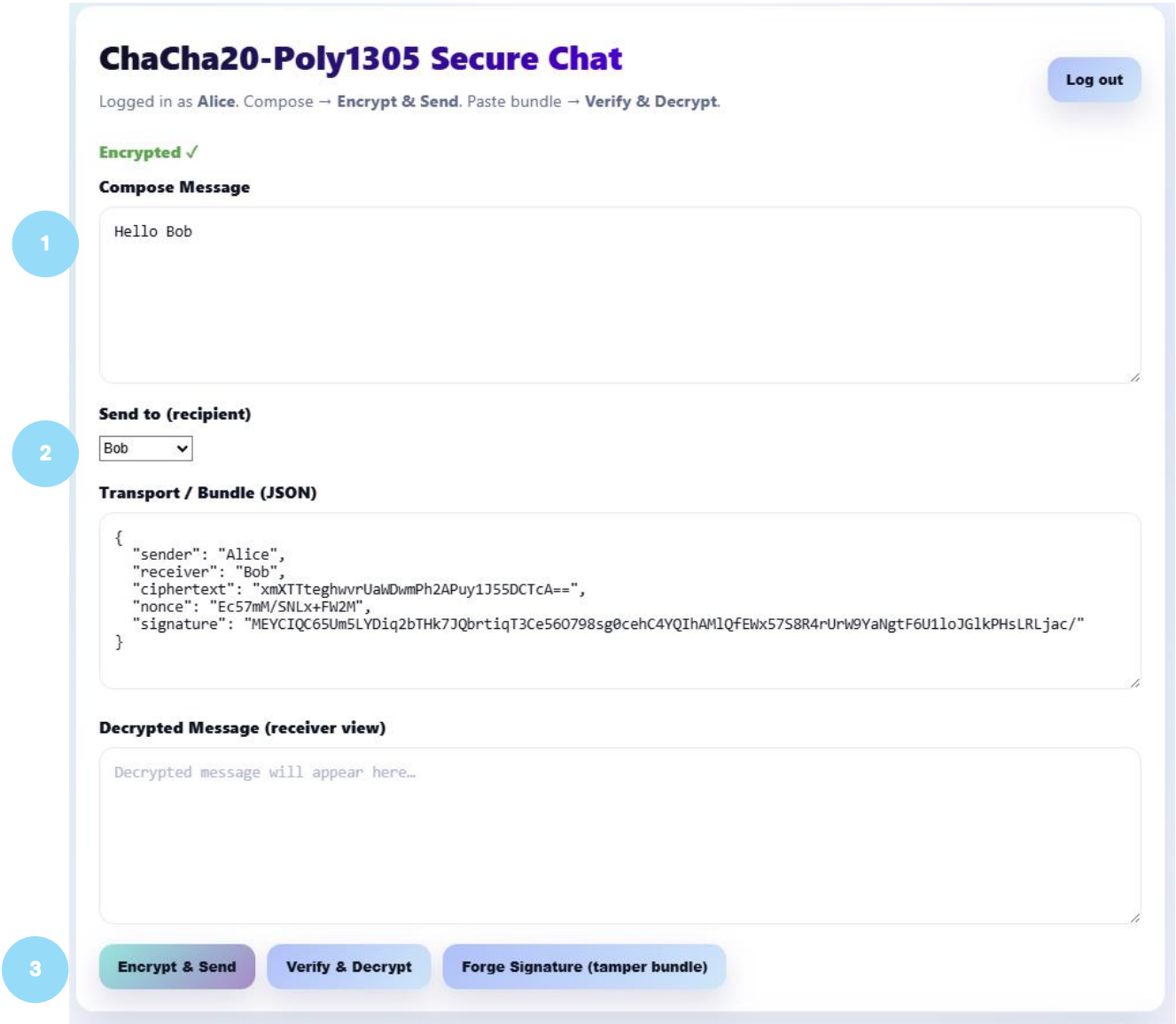
Example default test accounts:

- Alice / alice123
- Bob / bob123
- Charlie / charlie123



2. Sending a Message

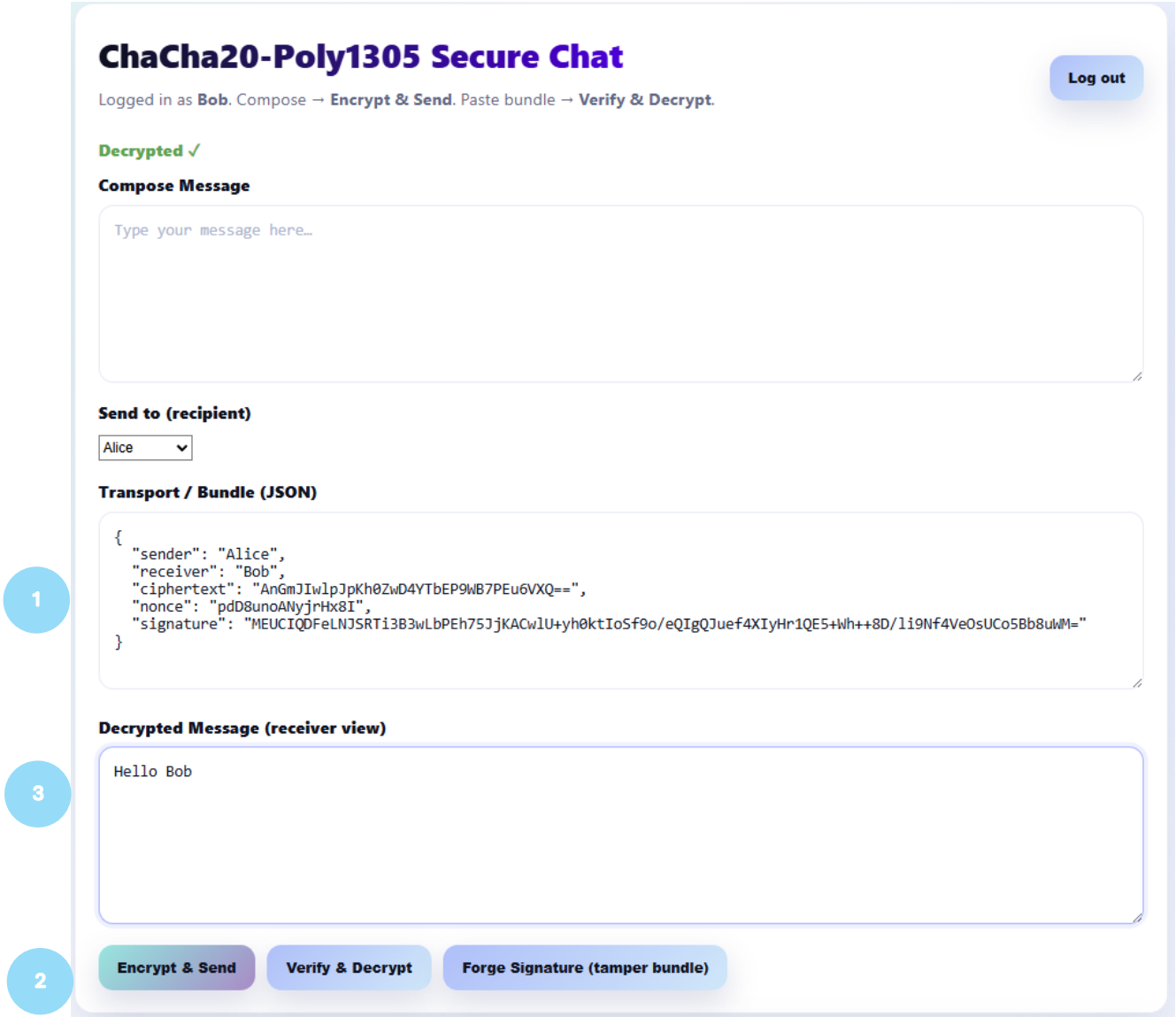
- Type your plaintext message in the message box
- Select the intended recipient from the dropdown list
- Press **Encrypt & Send**



3. Decrypting a Message

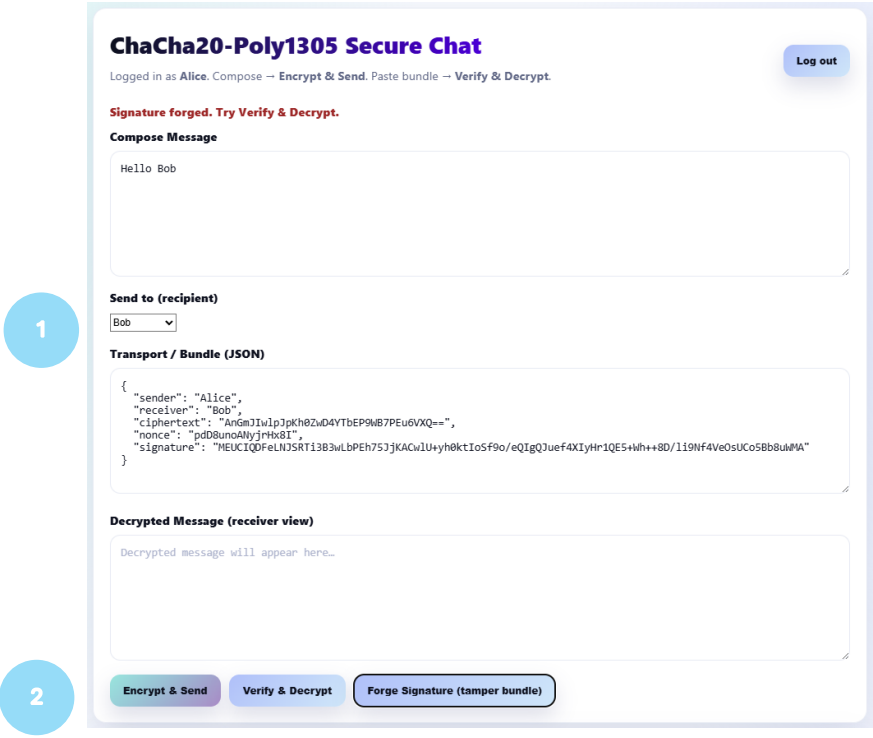
To simulate sending a message:

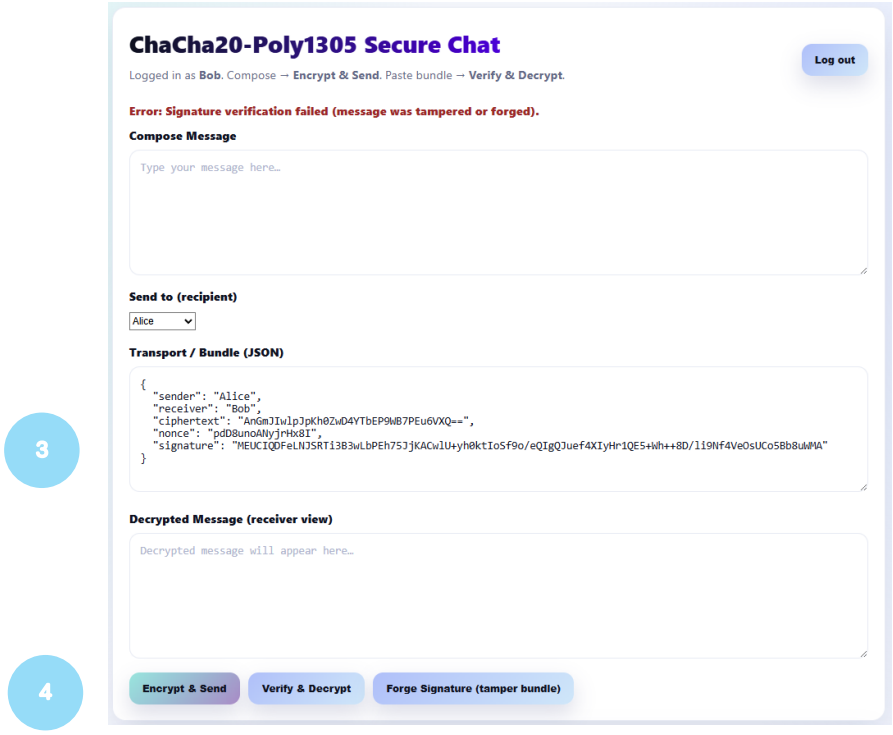
- Copy the generated Transport Bundle JSON
- Paste it into the second text box on the receiver's screen (second window)
- On the receiving side click "Verify & Decrypt"



4. Simulate Forged Signature

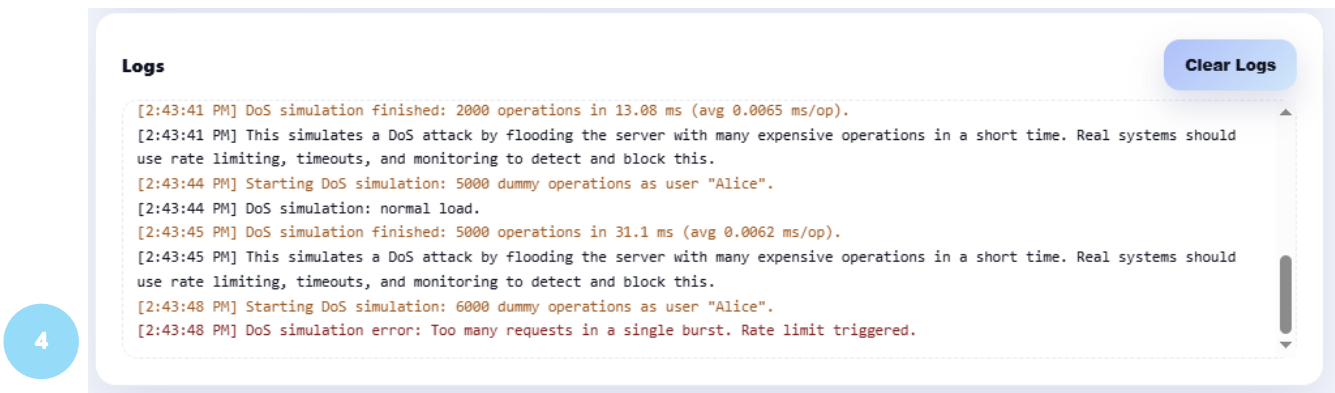
- Send a message normally (Step 2)
- Click “Forge Signature”
- Attempt to decrypt (Step 3)





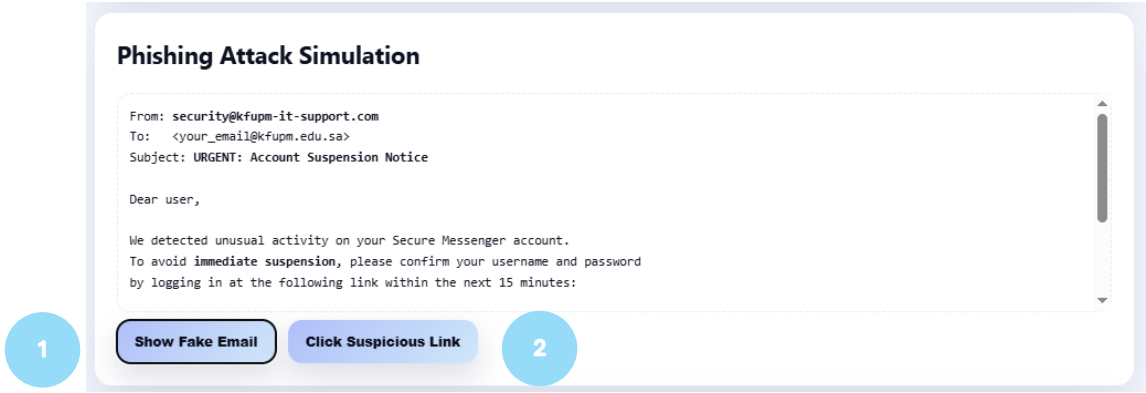
5. Simulate DoS Attack

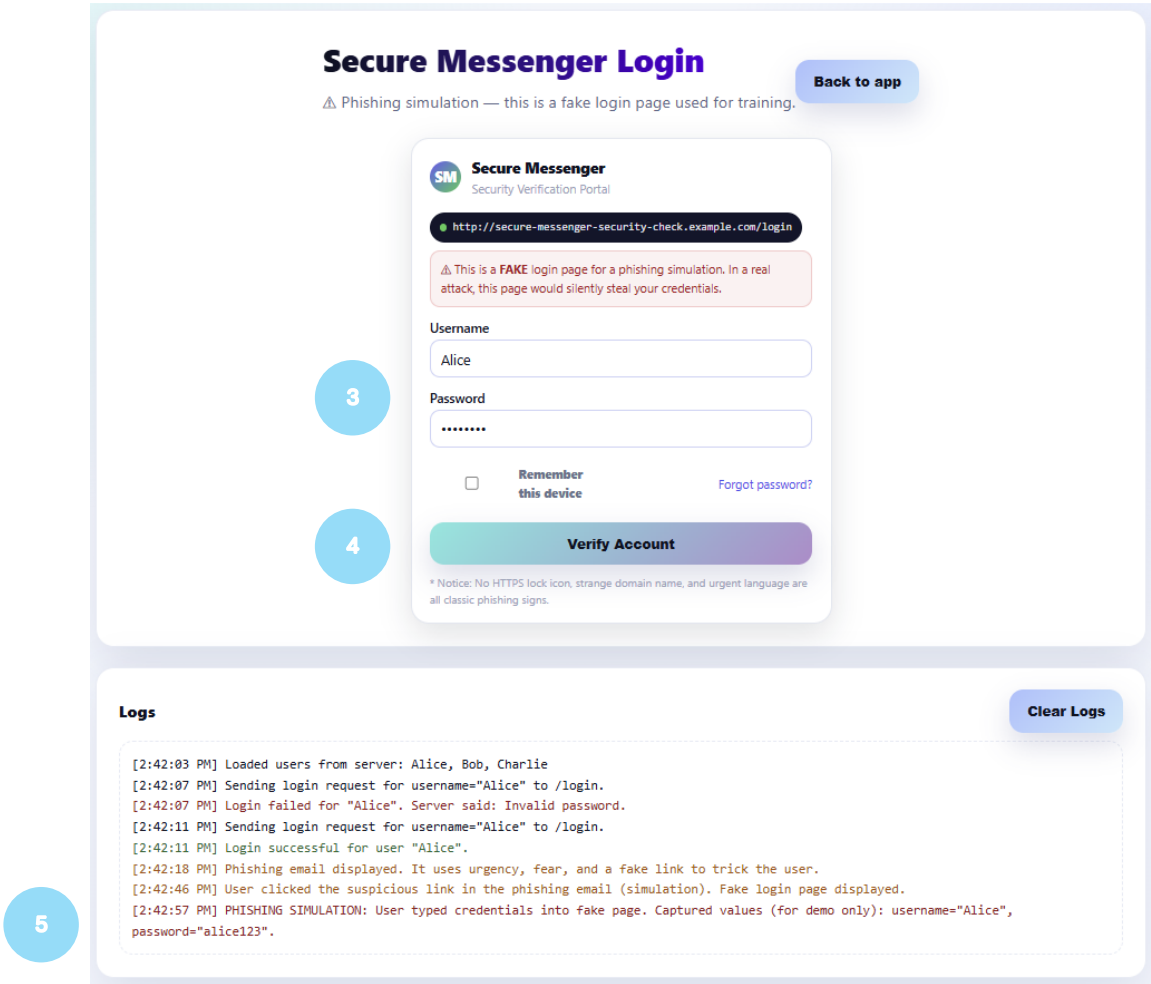
- Enter number of simulated requests
- Click “Simulate DoS Flood”
- Increase number of requests to view increase in resource usage
- Select > 5000 requests to trigger rate limit

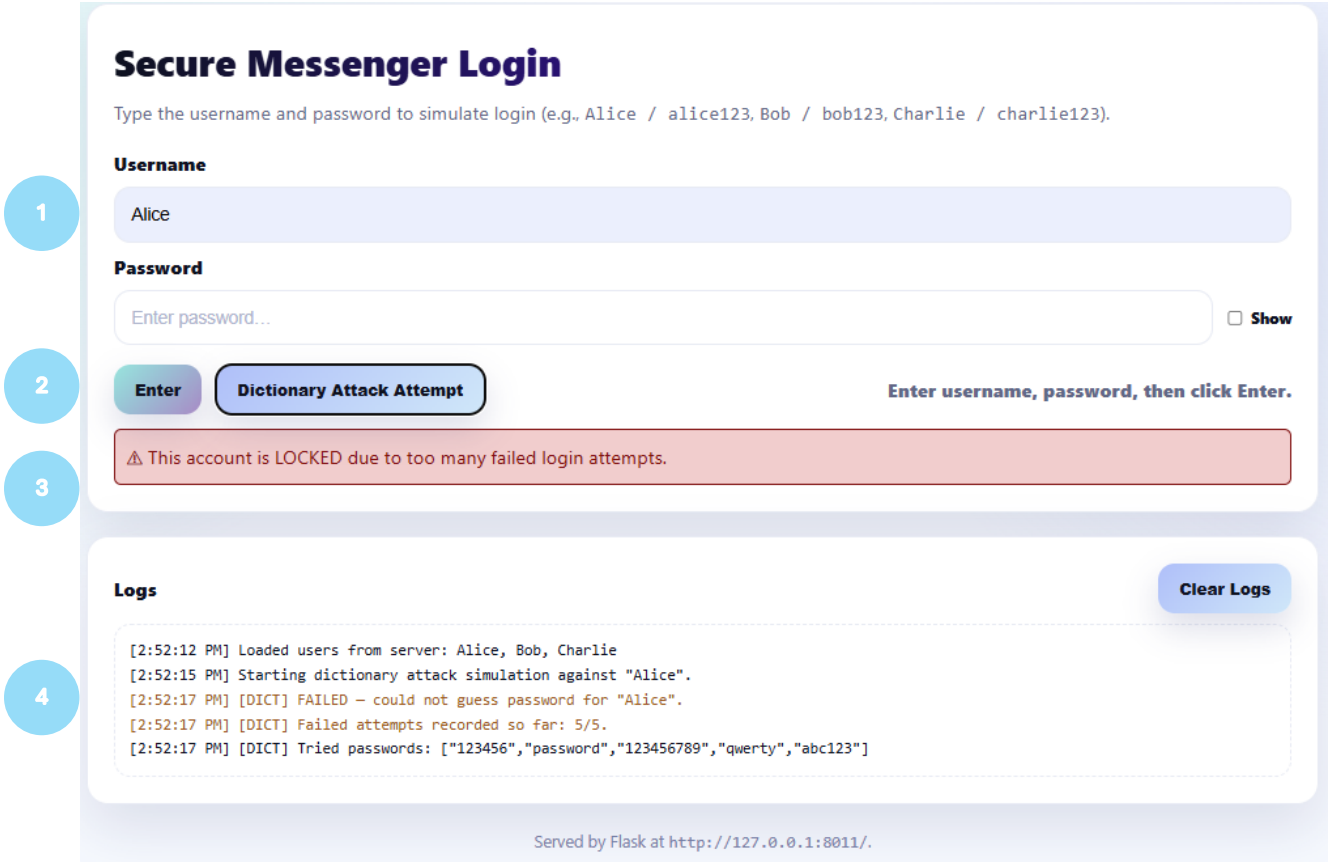


6. Simulate Phishing Attack

- Press “Show Fake Email”
- Click “Suspicious Link”
- Fake login appears
- Enter username and password
- View captured values in logs





7. Simulate Dictionary Attack

- Go to login page
- Enter username of target user
- Click “Dictionary Attack Attempt”